

One trusted ARR answer, ready for humans and AI

â€¢ A semantic layer on top of the existing data platform

â€¢ One governed metric, proven end to end

Why this project exists

â€¢ Boards and RevOps do not need more dashboards.

â€¢ They need one ARR answer that survives BI, finance review, and AI reuse.

â€¢ The case study is not a platform survey. It is a proof that one important metric can be defined, certified, and consumed safely.

What the project actually proves

- â€¢ A single end-to-end ARR path from synthetic Salesforce-style inputs to:
- â€¢ a certified month-end ARR fact
- â€¢ an explainable ARR movement fact
- â€¢ dbt semantic metrics
- â€¢ a Snowflake semantic-view consumption path
- â€¢ tests, contracts, and a repeatable demo runbook

The metric is the product

- â€¢ Ending ARR is not just a sum of active rows.

- â€¢ It is a governed business definition with:

- â€¢ month-end certification

- â€¢ effective-date logic

- â€¢ eligibility rules

- â€¢ annualization rules

- â€¢ approved dimensions

- â€¢ invalid uses

- â€¢ a change process

ARR rules that make the project non-trivial

â€¢ The MVP rules are explicit:

â€¢ calendar month-end only

â€¢ USD only

â€¢ annualize recurring lines with $\text{'net_amount_per_period'} \wedge 12 / \text{billing_interval_months}$

â€¢ exclude one-time fees, services, credits, and tax

â€¢ use effective dates, not just current status

â€¢ preserve historical inclusion for cancelled or expired subscriptions

Why the data model is layered this way

â€¢ Synthetic Salesforce CSV seeds feed staging views, then one intermediate ARR rule model, then certified marts, then semantic consumption.

â€¢ That separation matters:

â€¢ staging cleans names and types

â€¢ intermediate owns business logic

â€¢ marts expose certified facts

â€¢ semantic layers expose approved meaning

What the business logic model does

- â€¢ 'int_subscription_arr_lines' is the calculation boundary.

- â€¢ It joins subscription lines, subscriptions, and products to determine:

- â€¢ whether a line is ARR-eligible

- â€¢ how much ARR it contributes

- â€¢ whether the line should stay visible for auditability

- â€¢ Ineligible lines are not deleted. They remain inspectable with 'is_arr_eligible = false' and 'line_arr = 0'.

How we know the metric is correct

- â€¢ The repo does not rely on one validation method.
- â€¢ It uses a stack:
 - â€¢ hand-calculated expected totals
 - â€¢ singular business tests
 - â€¢ schema tests and model contracts
 - â€¢ unit tests for the complex logic
 - â€¢ semantic validation / contract checks
- â€¢ That is what makes the metric reviewable instead of merely runnable.

The edge cases are the point

- â€¢ The synthetic dataset was designed around real ARR scenarios:

- â€¢ monthly, quarterly, and annual billing

- â€¢ discounts

- â€¢ one-time and services exclusions

- â€¢ subscription start and end boundaries

- â€¢ churn and reactivation

- â€¢ renewal uplift

- â€¢ multiple eligible lines at one grain

- â€¢ This is what keeps the project from feeling like a toy example.

Ownership and change control

- The contract is as important as the SQL.

- RevOps owns the business meaning

- Data owns implementation and reliability

- changes require review, effective date handling, and re-certification

- That avoids a common failure mode: the metric changes quietly while the dashboard still looks correct.

Why the semantic layer matters

â€¢ The semantic layer is the control point that stops every consumer from rebuilding ARR differently.

â€¢ It is also where the project surfaces a real architectural tradeoff:

â€¢ dbt provides lineage, governance, and testing

â€¢ Snowflake semantic views provide a governed consumption path for BI and AI

â€¢ One certified definition should serve both.

AI should only see certified meaning

- â€¢ The AI story is intentionally constrained.

- â€¢ AI must read:

- â€¢ certified metrics only

- â€¢ approved dimensions only

- â€¢ lineage-aware objects only

- â€¢ verified queries or semantic surfaces only

- â€¢ The goal is not a free-form assistant over raw tables.

- â€¢ The goal is a controlled consumer of trusted business meaning.

What remains open by design

- â€¢ The project is strong, but it is still scoped as a lab.
- â€¢ Still deferred:
 - â€¢ live Metabase deployment
 - â€¢ live Snowflake semantic execution in this workspace
 - â€¢ recognized ARR and reconciliation
 - â€¢ multi-currency and historical type-2 account dimensions
 - â€¢ production ingestion through true warehouse sources
- â€¢ That scope boundary is intentional; it keeps the case study credible.

Success criteria

â€¢ The deck succeeds if a reviewer can say:

â€¢ I understand the ARR contract

â€¢ I can see how the model is validated

â€¢ I know who owns the metric

â€¢ I understand why the semantic layer is needed

â€¢ I can see how AI would safely consume it

â€¢ If the answer changes by surface, the model is not finished.

â€¢ The point is not to create another abstraction.

â€¢ The point is to create one shared business vocabulary that humans trust and AI can safely use.